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# The tendency to trust is reflected in human brain structure

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ABSTRACT

Trust is an important component of human social life. Within the brain, the function within a neural network implicated in interpersonal and social-cognitive processing is associated with the way trust-based decisions are made. However, it is currently unknown how localized structure within the healthy human brain is associated with the tendency to trust other people. This study was designed to test the prediction that individual differences in the tendency to trust are associated with regional gray matter volume within the ventromedial prefrontal cortex (vmPFC), amygdala and anterior insula. Behavioral and neuroimaging data were collected from a sample of 82 healthy participants. Individual differences in the tendency to trust were measured in two ways (self-report and behaviorally: trustworthiness evaluation of faces task). Voxel based morphometry analyses of high-resolution structural images (VBM8-DARTEL) were conducted to test for the association between the tendency to trust and regional gray matter volume. The results provide converging evidence that individuals characterized as trusting others more exhibit increased gray matter volume within the bilateral vmPFC and bilateral anterior insula. Greater right amygdala volume is associated with the tendency to rate faces as more trustworthy and distrustworthy (U-shaped function). A whole brain analysis also shows that the tendency to trust is reflected in the structure of dorsomedial prefrontal cortex. These findings advance neural models that associate the structure and function of the human brain with social decision-making and the tendency trust other people.



## Introduction

Trust is a valuable component of human social life. The ability to trust others strengthens social relationships and improves the efficiency of dynamic social groups. Recently, considerable progress in social-cognitive neuroscience research demonstrates that the tendency to trust is subserved by the function of specific brain regions (Adolphs, 2002; Dimoka, 2011; Krueger et al., 2007; Rilling and Sanfey, 2011). However, it is currently unknown if the tendency to trust is associated with variation in the structure of the healthy human brain. This study was designed to test the prediction that individual differences in the tendency to trust are associated with the gray matter volume in brain regions subserving interpersonal social-cognitive function.

The network of brain regions that subserve individual differences in the tendency to trust may include loci engaged during psychological tasks designed to measure trust explicitly. Trust tasks used in neuroimaging research often involve economic scenarios (such as the "trust game") (Berg et al., 1995; Delgado et al., 2005; van den Bos et al., 2009) or the evaluation of trustworthiness of social stimuli, such as facial expressions (Todorov, 2008; Todorov et al., 2008). Across different types of trust tasks, there exists a consistent pattern of results; trust is associated with increased neural activity within brain regions involved in social evaluation and interpersonal function (Riedl and Javor, 2012; Rilling and Sanfey, 2011) that includes the ventromedial prefrontal cortex (vmPFC), amygdala and anterior insula.

The vmPFC is involved in coding value and social decision making (Rushworth et al., 2011). vmPFC activity is often observed during personal and interpersonal decision making (Cloutier and Gyurovski, 2014; Kim and Johnson, 2014), as well as emotion regulation (Silvers et al., 2014). During trust tasks, vmPFC activity is associated with reciprocity and cooperation (Cooper et al., 2010; Rilling et al., 2002; Sakaiya et al., 2013) and dynamic changes in trustworthiness of facial expressions (Killgore et al., 2013). Damage to the vmPFC is associated with abnormal social decision making (Moretti et al., 2009) and with reduced trust-based decisions during the trust game (Krajbich et al., 2009; Moretto et al., 2013). These studies support the hypothesis that vmPFC structure is a neural construct associated with individual differences in the tendency to trust.

The amygdala functions to code the emotional salience of information (Aggleton, 2000). Across a wide range of tasks, amygdala activity increases during social and emotional evaluation (Costafreda et al., 2008). Functional neuroimaging research on trust shows that the amygdala is engaged during trustworthiness evaluation of faces (Engell et al., 2007; Winston et al., 2002), and that amygdala activity correlates with the subjective evaluation of trustworthiness of social stimuli (Rule et al., 2013). Rule and colleagues (2013) recently reported that

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amygdala activity increases when making judgments related to trustworthiness and distrustworthiness. Furthermore, individuals with amygdala damage display abnormal trustworthiness evaluations of faces (Adolphs et al., 1998) and altered trust based decision making (Koscik and Tranel, 2011). Together, these findings indicate that amygdala structure may be a neural construct associated with individual differences in the tendency to trust and distrust.

The anterior insula is involved in emotional awareness and the subjective experience of emotions (Gu et al., 2013). In terms of trust, anterior insula activity is associated with trustworthiness decisions of faces (Kragel et al., 2014; Winston et al., 2002) and with trust based decisions during economic scenarios (Aimone et al., 2014; van den Bos et al., 2009). Additionally, empirical evidence indicates that the anterior insula codes both trust (Killgore et al., 2013) and distrust (Winston et al., 2002) evaluations of faces. Together, these findings support the hypothesis that anterior insula structure may correspond to individual differences in the tendency to trust.

In this study, we measured individual differences in the tendency to trust in two ways (self-report and behaviorally) and collected high-resolution structural brain imaging data from of healthy population ($n = 82$). We used a self-report measure of trust, consisting of the A1: Trust facet of the personality trait agreeableness within the NEO-PI-3. Higher scores on the Trust facet reflect a disposition to believe that other people are honest and well intentioned (Costa et al., 1991). For the behavioral measure of trust, we used a trustworthiness evaluation of faces task (Rule et al., 2013; Todorov, 2008). We used VBM8-DARTEL to quantify gray matter volume using a whole brain approach and within *a priori* regions of interest (vmPFC, amygdala and anterior insula). Based on existing functional neuroimaging evidence, we predicted that individual differences in the tendency to trust (self-report and behaviorally) would be associated with gray matter volume within the vmPFC, amygdala and anterior insula.

## Methods

### Participants

We recruited 82, fluent English-speaking (49 females, 33 males; mean age = 20.74 years, standard deviation = 3.02 years, 80 right-handed) adults from the University of Georgia and surrounding community to participate in a study on brain function and social behavior. All participants were screened for neurological conditions and MRI counter indications. All participants provided written informed consent as detailed in the Declaration of Helsinki, and the University of Georgia Institutional Review Board approved all procedures within this study.

### Tendency to trust: Self-report

Each participant completed the NEO Personality Inventory-3 (NEO PI-3). The NEO PI-3 covers each of the Big 5 personality traits (neuroticism, extraversion, openness, agreeableness, and conscientiousness) and includes the A1: Trust facet for agreeableness. The Trust facet is designed to characterize the tendency to attribute benevolent intent to others and the disposition to believe that others are honest and well intentioned (Costa et al., 1991). An example of an item from the Trust facet is, “My first reaction is to trust people.” Empirical investigations show that the Trust facet has high internal consistency (Costa and McCrae, 1995; Costa et al., 1991) and is a valid metric to quantify the biological basis of the tendency to trust (Cardoso et al., 2013). Prior psychometric analyses of the A1: Trust facet of agreeableness shows high convergent and discriminant validity (Costa et al., 1991). Data were scored to represent *T*-values, with the population mean defined as $T = 50$ and one standard deviation of $T = 10$. The sample scores for Trust ($M = 46.85$, SD = 11.99) were within the range of the normal non-clinical population. Reliability analysis for all items within the Trust facet showed high internal consistency (Chronbach's $\alpha = .81$).

### Tendency to trust: Behavior

Each participant completed a behavioral task designed to measure trustworthiness evaluation of faces (Rule et al., 2013; Todorov, 2008). Each participant was presented with thirty-six photographs of people displaying neutral facial expressions, selected from a standardized database of face stimuli (Minear and Park, 2004). Of the thirty-six photographs, half (18) were female and half were male. Each participant was instructed to evaluate each face, based on how “trustworthy the person in the photograph appeared to be,” on a seven point Likert scale, with 1 being “distrust” and 7 being “trust.” Mean trustworthiness evaluations were calculated for each participant. Reliability analysis for all of the faces rated on trustworthiness showed high internal consistency (Chronbach's $\alpha = .94$). Mean trustworthiness evaluation of faces values did not differ according to sex or age of participants ($p > .10$).

### MRI data acquisition

High-resolution whole-brain imaging data using a T1-weighted spoiled grass gradient recalled sequence were acquired on a GE-Signa 3.0 T scanner (General Electric, Milwaukee, WI) at the University of Georgia Bio-imaging Research Center (UGA BIRC: http://birc.uga.edu/). The following parameters were used: repetition time, 24 ms; echo time, 4.5 ms; flip angle, 20°; matrix size, $256 \times 256$; field of view, 25.6 cm; slice thickness, 1.0 mm; 164 contiguous slices.

### MRI data preprocessing

Images were initially visually inspected for artifacts or structural abnormalities. VBM analyses were performed using SPM8 (http://www.fil.ion.ucl.ac.uk/spm/software/spm8/) as implemented through the VBM8 toolbox. First, the origin of each participant's structural image was manually set to the anterior commissure. Next, each image was segmented into gray matter, white matter, and cerebrospinal fluid, and then transformed to MNI stereotactic space using affine and non-linear spatial normalization. The segmented images were then iteratively registered by the Diffeomorphic Anatomical Registration Through Exponentiated Lie algebra toolbox (Ashburner and Friston, 2000). This process created a template for the group of individuals. The resulting template image was transformed to MNI stereotactic space using affine and non-linear spatial normalization with intensity modulation by the Jacobian determinant of the deformation flow field computed for each image. The resulting images were transformed to MNI space and smoothed with an isotropic Gaussian kernel of 10 mm full width at half maximum.

### Voxel based morphometry (VBM) analysis

We used a multiple regression analysis, within SPM8 software, to investigate the association between each trust metric (self-report and behavioral) and gray matter volume. For all analyses, age, sex and total cranial volume were entered as covariates. First we examined the association between self-report (Trust facet) scores and regional gray volume. Next, we examined the association between mean trustworthiness evaluation of faces values (behavioral measure) and regional gray matter volume. Lastly, we investigated the association between both trust measures and regional gray matter volume by performing a whole brain analysis with both trust measures entered as independent variables. To protect against false positive (Type 1) errors, we used a combined non-parametric and parametric statistical approach. First, we used a non-parametric approach to create an inclusive mask for areas that survived a whole brain (FDR-corrected, $p < .05$) analysis, using threshold free cluster enhancement (TFCE) (Smith and Nichols, 2009). The TFCE analysis was implemented with the TFCE toolbox in SPM8. For TFCE, we calculated cluster level statistics (correlations with each trust measure) based on a total 5000 permutations. The TFCE analysis was performed on data smoothed using a 2 mm full width at half maximum kernel. Next, we used a parametric approach (restricted to areas within the non-parametric, TFCE, FDR-corrected mask), using a combined spatial extent and height threshold of 172 voxels and $p < .005$ for each cluster based on a prior VBM analysis using similar parameters (Lu et al., 2014). For whole brain analysis, we only report clusters that survived following the non-parametric and parametric statistical procedures (Silver et al., 2011).

For the amygdala region of interest, we extracted data using a predefined mask selected from a standardized atlas used for functional and structural neuroimaging (Maldjian et al., 2003) restricted to areas surviving whole brain FDR $p = .05$ corrected TFCE analysis (Smith and Nichols, 2009). Based on evidence that the amygdala is involved in the assessment of distrust and trust (Rule et al., 2013; Todorov, 2008), we performed two types of analyses. For the amygdala, we tested for linear associations (with each trust metric) and quadratic associations (U- and inverted U-shaped function) with each trust metric. Lastly, based on evidence that that insula is also associated with both trust (Killgore et al., 2013) and distrust (Winston et al., 2002) evaluations of faces, we also used tested for a quadratic associations within the insula using an atlas based mask (Maldjian et al., 2003).

## Results

### *Trust measures*

Bivariate correlation analysis indicated that Trust facet scores were positively associated with mean trustworthiness evaluations of faces values ($r = .44$, $p < .001$). The association between the Trust facet scores and mean trustworthiness evaluations of faces values remained statistically significant when age and sex were entered as covariates ($r = .43$, $p < .001$). These findings indicate convergent validity across each measure of trust. Self-report and the behavioral measure of trust were not statistically associated with global measures of brain volume; including total, gray or white volume, or cerebrospinal fluid volume.

### *Trust self-report and gray matter volume*

Results of whole brain analysis with Trust facet scores predicting gray matter volume are presented in Table 1 and Fig. 1. Within Fig. 1, areas exhibiting a significant association with Trust facet scores are shown in green. Trust facet scores were positively associated with gray matter volume within the bilateral ventromedial prefrontal cortex (MNI: 4, 46, −5; $t = 3.76$, $p < .001$) and right anterior insula (MNI: 40, 16, −15; $t = 4.13$, $p < .001$). Trust facet scores were negatively associated with gray matter volume within the bilateral cerebellum.

### *Trustworthiness evaluations and gray matter volume*

Results of whole brain analysis with mean trustworthiness evaluation of faces values predicting gray matter volume are presented in Table 1 and Fig. 1. Within Fig. 1, areas exhibiting a significant association with mean trustworthiness evaluation of faces values are shown in violet. Mean trustworthiness evaluation of faces values were positively associated with gray matter volume within the bilateral ventromedial prefrontal cortex (MNI: −14, 39, −9; $t = 4.10$, $p < .001$) and bilateral anterior insula (left: MNI: −30, 30, −5; $t = 3.48$, $p = .001$, right: MNI: 40, 15, −2; $t = 2.95$, $p = .002$). We did not observe that mean trustworthiness evaluation

**Table 1**
Anatomical regions displaying significant association with each trust metric.

| Anatomical region | L/R | MNI *x* | *y* | *z* | *t* | *k* |
|---|---|---|---|---|---|---|
| *A1 Trust (self-report): positive correlation* | | | | | | |
| Inferior temporal gyrus | R | 66 | −33 | −26 | 4.65 | 3043 |
| Dorsomedial prefrontal cortex | Bilateral | 4 | 23 | 40 | 4.65 | |
| Extending into: | | | | | | |
| Ventromedial prefrontal cortex | Bilateral | 4 | 46 | −5 | 3.76 | |
| Lateral orbitofrontal gyrus | L | −24 | 50 | −12 | 3.72 | 16031 |
| Anterior insula | R | 40 | 16 | −15 | 4.13 | 2089 |
| Superior frontal gyrus | L | −28 | 35 | 43 | 4.11 | 2159 |
| Superior frontal gyrus | R | 34 | 8 | 46 | 4.04 | 1275 |
| Precentral gyrus | L | −60 | −9 | 42 | 3.53 | 265 |
| Cerebellum | L | −39 | −39 | −36 | 3.45 | 804 |
| Inferior frontal gyrus | R | 50 | 32 | 0 | 3.31 | 786 |
| Precentral gyrus | R | 50 | −9 | 43 | 3.17 | 284 |
| Inferior temporal gyrus | L | −60 | −25 | −18 | 2.90 | 195 |
| Thalamus | R | 2 | −7 | 7 | 2.82 | 271 |
| *A1 Trust (self-report): negative correlation* | | | | | | |
| Cerebellum | Bilateral | 3 | −49 | −29 | 3.76 | 1129 |
| Cerebellum | L | −16 | −37 | −44 | 3.49 | 188 |
| *Trustworthiness evaluation of faces: positive correlation* | | | | | | |
| Ventromedial prefrontal cortex | Bilateral | −14 | 39 | −9 | 4.10 | 2335 |
| Anterior insula | L | −30 | 30 | −5 | 3.48 | 832 |
| Dorsomedial prefrontal cortex | Bilateral | −2 | 29 | 61 | 3.40 | 181 |
| Anterior insula | R | 40 | 15 | −2 | 2.95 | 344 |
| *Trustworthiness evaluation of faces: negative correlation* | | | | | | |
| Cerebellum | Bilateral | 6 | −52 | −29 | 3.94 | 1796 |
| Middle occipital gyrus | R | 38 | −66 | 0 | 3.68 | 588 |
| Conjunction: positive correlation | | | | | | |
| Ventromedial prefrontal cortex | Bilateral | −14 | 39 | −9 | 4.10 | 1061 |
| *Conjunction: negative correlation* | | | | | | |
| Cerebellum | Bilateral | 6 | −52 | −29 | 3.94 | 586 |

Results reported for whole brain ($p = .05$, TFCE-FDR corrected threshold) with age, sex and total cranial volume entered as covariates; R = right; L = left.

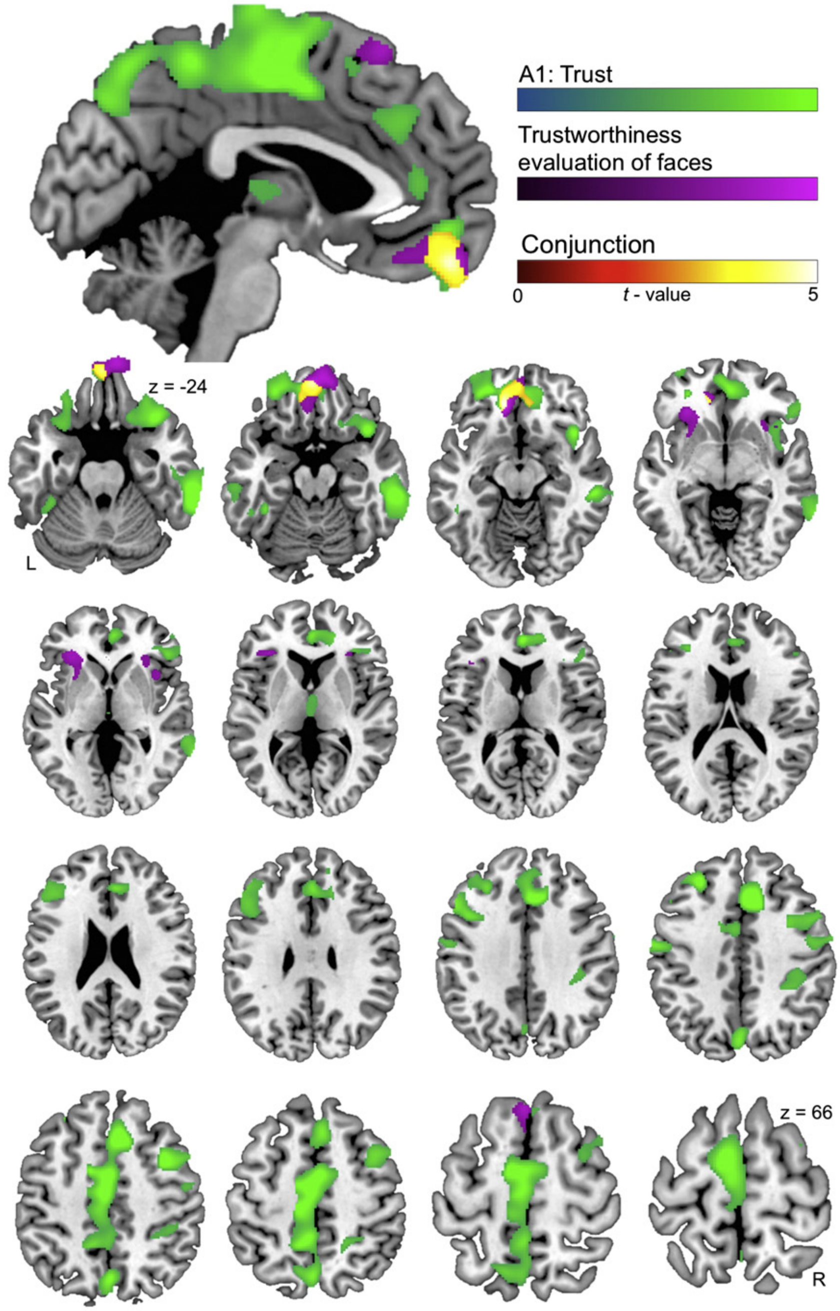


**Fig. 1.** Whole brain analysis associating the tendency to trust with regional gray matter volume. Regional gray matter volume differences are overlaid on a sagittal slice and axial slices of a standardized template brain. Brain areas exhibiting a significant association with Trust facet scores are shown in green. Brain areas exhibiting a significant association with mean trustworthiness evaluation of faces values are shown in violet. Brain areas exhibiting a significant association with both trust measures (conjunction: Trust facet scores and mean trustworthiness evaluation of faces values) are shown in yellow. Coordinates are in MNI space. R = right; L = left.

of faces values were linearly associated with amygdala volume. However, we observed that right amygdala volume was associated with a U shape function of mean trustworthiness evaluation of faces values (Fig. 2) (MNI: 30, 4, −29; 34 voxels; $t = 2.98$, $p = .002$). Specifically, we found that both the tendency to rate faces as more trustworthy and as more distrustworthy was associated with greater right amygdala volume. We did not observe that mean trustworthiness evaluation of faces values exhibited a quadratic association with anterior insula volume. The only brain region where a negative association with mean trustworthiness evaluation of faces values was observed was the bilateral cerebellum (MNI: 6, −52, −29; $t = 3.94$, $p < .001$).

*Conjunction analysis: Trust facet and trustworthiness evaluations*

Results of whole brain analysis with both Trust facet scores and mean trustworthiness evaluation of faces values predicting gray matter volume are presented in Table 1 and Fig. 1 (presented in yellow). We observed that both Trust facet scores and mean trustworthiness evaluation of faces values were positively associated gray matter volume within the bilateral vmPFC (MNI: −14, 39, −9; $t = 4.10$, $p < .001$). For the negative correlation, we observed that both trust measures combined, were negatively associated with gray matter volume within the bilateral cerebellum (MNI: 4, −52, −29; $t = 3.94$, $p < .001$).

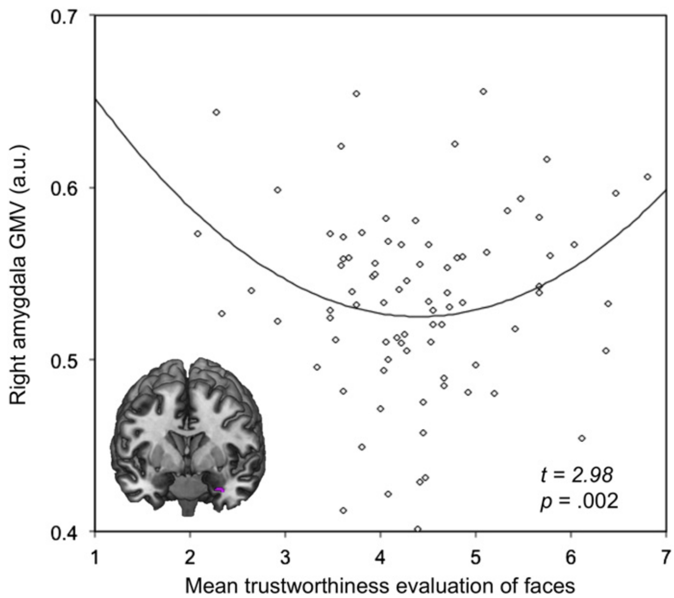


**Fig. 2.** Scatterplot showing the association between mean trustworthiness (U-shaped, quadratic function) of faces values and gray matter volume within the right amygdala (MNI: 30, 4, −29). Mean trustworthiness evaluation of faces values are plotted on the *x*-axis. Right amygdala gray matter volume estimates in arbitrary unites (a.u.) are plotted on the *y*-axis.

## Discussion

In this study, we demonstrate an association between individual differences in the tendency to trust and gray matter volume within a neural network implicated in interpersonal function. Heightened tendencies to trust, as measured by self-report and behaviorally (trustworthiness evaluation of faces), were associated with increased gray matter volume within the vmPFC and anterior insula. Additionally, we found that the tendency to evaluate face as more distrustworthy and more trustworthy was associated with greater gray matter volume within the right amygdala. These findings indicate that localized gray matter volume in the vmPFC, amygdala and anterior insula is associated with a disposition to trust others.

We found converging evidence that individuals characterized as being more trusting exhibit greater vmPFC gray matter volume compared to those tending to be less trusting. Increased vmPFC gray matter volume was observed in individuals scoring higher on the Trust facet and in those that tended to evaluate faces as more trustworthy. This finding strengthens existing neural models that characterize the vmPFC to be involved in coding social value (Behrens et al., 2008; Bzdok et al., 2013; Cloutier and Gyurovski, 2014; van den Bos et al., 2013). For example, Cloutier and Gyurovski (2014) recently used a functional neuroimaging approach to show preferential vmPFC activity in response to faces of varying social status (moral and financial). The vmPFC is active during tasks that specifically involve trust-based decision-making (Cooper et al., 2010; Rilling et al., 2002; Sakaiya et al., 2013). Furthermore, focal vmPFC lesions impact trust-based decision-making; individuals with vmPFC damage exhibit reduced reciprocity behavior as compared to controls (Moretto et al., 2013). Together, the converging evidence from the current study, and existing fMRI and lesion-based research, strongly support the role of the vmPFC in coding social value and trust-based decision-making.

We found that both measures of trust were associated with greater gray matter volume within the anterior insula. Existing models describing the function of the anterior insula include the processing of affective awareness and the somatic experience of emotions (Gu et al., 2013; Mutschler et al., 2009). For example, many studies show that anterior insula activity increases during the experience of negatively valenced emotional states including disgust (Chapman and Anderson, 2012) and anxiety (Damsa et al., 2009) and during the assessment of potential risks (Bossaerts, 2010). In terms of interpersonal function, anterior insula activity occurs during the violation of social norms as in unreciprocated cooperation (Rilling et al., 2008) and unfairness (Guo et al., 2014). Additionally, the anterior insula is involved in social decision-making. Hollmann et al. (2011) used real-time fMRI and a pattern classification approach to demonstrate that anterior insula activity can serve to predict social decisions during the ultimatum game. Tasks that explicitly involve trust are also associated with increased anterior insula activity (Aimone et al., 2014; van den Bos et al., 2009). Furthermore, Kang et al. (2010) showed that anterior insula activity mediates the effect of physical pain (cold temperature) on trust-based decision-making. We found that the insula was associated with the tendency to trust using a linear function, but not a quadratic function (as with the amygdala). This finding may indicate that the anterior insula serves to evaluate social stimuli and subsequently influence decision based on subjective emotional evaluation coded by the amygdala. This theory is consistent with anatomical research demonstrating that the amygdala maintains projections with the anterior insula (Flynn, 1999; Gu et al., 2013). Combined, increased anterior insula gray matter volume in those that tend to be more trusting may be a neural construct influencing the way trust and distrust concepts are subjectively valued as potential "social risks," which therefore impacts the way social decisions are made.

We found that increased amygdala gray matter volume was associated with lower and higher mean trustworthiness evaluations of faces. We did not observe an association between amygdala gray matter volume and self-reported tendency to trust (Trust facet). The specificity of this association may in part represent the characteristics of each trust measure used in this study. The trustworthiness evaluation of faces task may rely, to a greater extent, on emotional attributes of trust, while the self-report measure may rely more on cognitive attributes of trust. This dissociation is supported by the large body of evidence that the amygdala is consistently involved during the social evaluation of faces (Engell et al., 2007; Mende-Siedlecki et al., 2013; Rule et al., 2013; Winston et al., 2002). In addition, this is supported by the observation that amygdala volume was associated with both the tendency to evaluate faces as more trustworthy and distrustworthy, as each represent emotional saliency greater than neutral (Rule et al., 2013). However, there is also evidence that the amygdala is involved during trust-related tasks that involve economic scenarios, such as during the trust game (Koscik and Tranel, 2011). Combined, the current findings support that amygdala structure is associated with the assessment of emotional saliency and thus affects trust-based decision-making.

The results of the whole brain analysis also showed that the tendency to trust was associated with gray matter volume within several other brain regions implicated in social-cognitive and interpersonal function. For example, across both measures of trust, individual differences in the tendency to trust were associated with greater gray matter volume within the dorsomedial prefrontal cortex (dmPFC). Functionally, the dmPFC is engaged during self-referential processing (Gusnard et al., 2001; Northoff and Bermpohl, 2004), empathy (Bernhardt and Singer, 2012) and the formation of impressions of others (Mitchell et al., 2005). There is also evidence that the dmPFC is engaged during tasks that involve trust explicitly. Baron et al. (2011) showed that dmPFC activity mediates the way the amygdala learns impressions of trustworthiness. Interestingly, Lewis and colleagues (2012) used a VBM approach and showed that individual differences in a related construct, fairness, are associated with greater dmPFC gray matter volume. Combined, the association between the tendency to trust and dmPFC structure may in part represent the way trust concepts are internalized to the self and trust impressions are learned.

Although this study provides substantial new information about the neural basis of trust, it is also limited in several important ways. We measured the tendency to trust using a self-report and behavioral metric. Considerable progress has been made in trust-based research that involves social groups. For example, modified versions of the prisoner's dilemma task allow for the impact of social context on trust-based decision-making to be quantified (Declerck et al., 2010, 2014). Several of the brain regions found to covary with the tendency to trust are implicated in social decision-making specifically. This finding supports the hypothesis that gray matter volume may impact trust-based decision-making differently according to social context. Additionally, we used a neuroimaging approach focused on characterizing gray matter only. A more complete understanding of the brain basis of trust will be achieved with the inclusion of other neuroimaging modalities such as diffusion tensor imaging. Lastly, this study is limited in terms of the sample size. The statistical power to elucidate associations between the tendency to trust and gray matter volume will be improved by carrying out studies with larger sample sizes.

In conclusion, trust is a valuable component of human social life. The way people trust each other impacts social relationships and is integral to maintaining a sense of community within society. Studying the neural basis of trust will improve the understanding of psychiatric conditions characterized by abnormalities in social cognition and as associated with genetic variation, as within the oxytocin system (Haas et al., 2013). Within the human brain, greater volume within the vmPFC, amygdala and anterior insula is associated with a disposition towards trusting other people more.

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